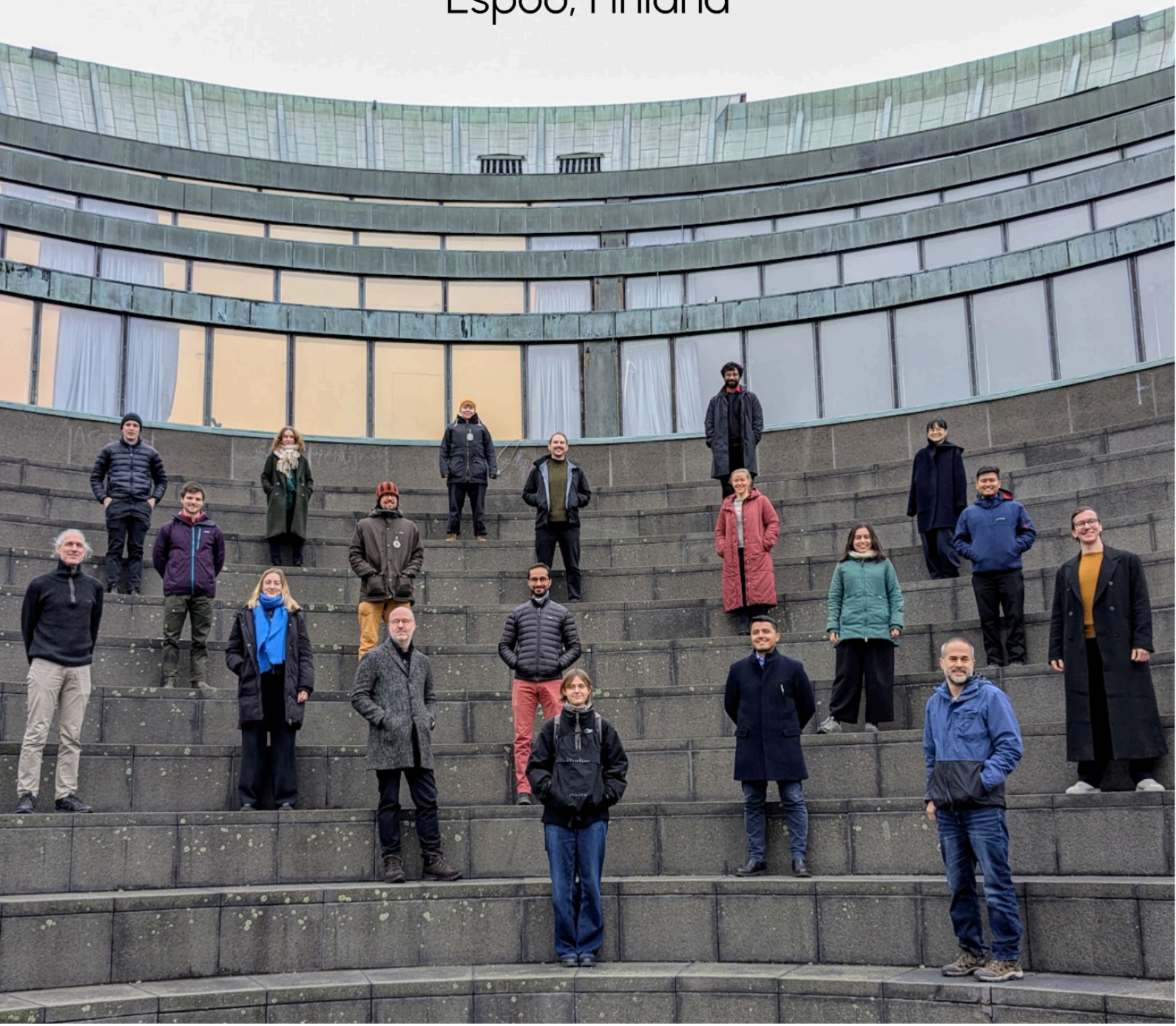




Global Freshwater Systems Science Workshop

30-31 January 2025

Aalto University
Espoo, Finland



Workshop schedule

Thursday, 30 January

9 –	Welcome from Xander & Vili	
10 –	Sharing update slides pt.1	<i>Xander & Vili to "mind-map" themes for breakout meetings</i>
	coffee break	
11 –	Sharing update slides pt.2	
12 –	Lunch @ Mau-kas	
13 –		
14 –	Collective discussion and work on position paper	
15 –	coffee break	
16 –	Breakout meeting <i>theme determined in morning</i>	Breakout meeting <i>theme determined in morning</i>
17 –	Dinner at Sen Viet @ 18:30	

note: we have 4 rooms booked, so can accommodate more mtgs. if needed

Friday, 31 January

9 –	Recap of day 1 & survey exercise for grp. paper		
10 –	coffee break		
11 –	Breakout working on position paper	Breakout meeting <i>new theme or continued from Thursday</i>	<i>note: we have 4 rooms booked, so can accomodate more mtgs. if needed</i>
12 –			
13 –	Lunch @ Mau-kas		
14 –	PhD student network	Discussion on collective's goals/needs	
15 –	coffee break		
	Final remarks and closing		

Participants

Lauren Seaby Andersen, Ronit Bohra, Heindriken Dahlmann, Simon Fahrländer, Ingo Fetzer, Dieter Gerten, Xander Huggins, Fernando Jaramillo, Chelsea Kaansdorp, Matti Kumm, David Lindao, Romi Lotcheris, Elina Paavonen, Miina Porkka, Juan Rocha, Lan Wang-Erlandsson, Petr Vesnovskii, Bhattarabhop Viriyaroj, Vili Virkki

Pre-workshop instructions:

Prepare a slide deck consisting of 2-3 slides on recent and planned future work

- **(optional)** Slide 1: recent research output/news to share
- Slide 2: actively ongoing research
- Slide 3: big picture / unanswered question or interest related to large-scale freshwater resilience (e.g. what kind of questions can this collective help explore and think through)

Please send your slide decks to Xander (xander.huggins@ubc.ca) and Vili (vili.virkki@uef.fi) at latest on **January 29th**.

Day 1 – January 30th – Aalto University Learning Centre (Otaniementie 9)

9:00–12:00 with a break

Room R015/213 Eetu

- Introduction to workshop agenda
- Collective member updates by brief slide decks (*short presentations*)
- Xander & Vili to summarise research themes and map conceptual connections on white board at the end of session
- Form potential research thematic breakout groups

12:00–13:30 – Lunch at [Mau-kas](#)

13:30–15:15

Room R015/213 Eetu

- Overview and discussion on our [position paper draft](#)
- Xander & Vili to introduce the draft and facilitate discussion on how to refine it
- One possible aim: which kind of data (orientation axes) to be collected for comparing different approaches and frameworks
- Xander & Vili to prepare a trial sheet for collecting data

15:30–17:00

Rooms R015/213 Eetu, R015/107 Arttu, R015/108 Jaakko, R015/225 Eeva

- Research thematic breakout groups

18:30 – Dinner at [Sen Viet](#) (Atlantinkatu 18)

- Walking from Ruoholahti metro station takes [15–20 minutes](#) (or [tram 8](#))
- [Tram 9](#) goes directly to the city centre from the restaurant (~15 minutes)

Day 2 – January 31st – Aalto University Learning Centre (Otaniementie 9)

9:00–10:15

Room R015/213 Eetu

- Recap of day 1 and feedback for day 2
- Sketching system connections for water resilience case studies

10:30–12:30

Rooms R015/213 Eetu, R015/107 Arttu, R015/108 Jaakko, R015/225 Eeva

- Breakout group for writing the position paper or engaging with it in some other, in-depth ways (Vili and/or Xander to host this)
- Research thematic breakout groups and/or continuations of day 1 discussions
- Group photo

12:30–14:00 – Lunch at [Mau-kas](#)

14:00–15:00

Rooms R015/213 Eetu, R015/107 Arttu, R015/108 Jaakko

- Breakout group for PhD student networking / peer support / needs and ambitions (PhD students only)
- Breakout group for sustaining the collective; e.g. group name, [mission statement](#), freshwater resilience conference session proposal, next workshop organisation and themes, possible funding, ambitions to grow or stay similar, collective website etc.

15:00 – Final remarks

Room R015/213 Eetu

- Final remarks and closing of workshop

Workshop recap

1. Position paper status and development

We discussed how freshwater systems can be seen as social-ecological systems and what might be the new openings, utility, or scientific advances/interest coming from this perspective. We debated how much we want to frame our narrative by offering solutions or solving specific issues, especially if we somehow state that our approach “empowers” something else. Less abstract terminology was desired to make our argumentation more approachable, and we considered leaving “freshwater resilience” out of the core framing in favour of this becoming more of a point of discussion. However, the key motivations of the study stemming from challenges of defining common language, scales and system boundaries in interconnected freshwater systems were well received.

In practical terms, we discussed which existing frameworks could be used in supporting and illustrating our argument that specialized frameworks suffer from disciplinary biases (e.g., system boundaries), and also what barriers hinder working across disciplines/frameworks. Furthermore, we listed some thematic

positioning axes to characterize different frameworks. We discussed how case studies would help to illustrate the applicability of our approach in, for instance, finding where systemic knowledge is stronger or weaker. It was requested that paper contributions should be clarified with leaders primarily responsible for this.

2. Selecting impactful case studies

We did a paper exercise to brainstorm on which kind of new insights our position paper could bring and which kind of case studies could be investigated to support our argumentation. The key purpose of our case studies would be to provide more detailed/tangible insights on system boundaries and connections that link each case to broader contexts, including global. We tentatively selected the Amazon rainforest, the Mekong, California, and Sahel as case studies since they are highly interconnected systems in which freshwater has a major role and they comprise a wide variety of different ecological, social, and governance challenges.

We drew on whiteboard connections between different case-specific SES elements to try out how each case could be conceptualized through the SES lens. Furthermore, we discussed state variables that could describe the state and functions of each case-specific interconnected system. These variables may help in illustrating the interactions and dependencies within and between local and global scales and make our central argument of adopting a SES perspective to freshwater systems science more approachable.

3. Discussion on workshop organization

We discussed whether we should label ourselves as a “collective”, “workshop”, or “group”, and set a short-term goal of creating a web page listing our members and activities. We may apply for funding to host workshops, estimating around €10–15k per workshop if held at an external venue. We will continue hosting calls to develop the position paper and establish an idea board and wishlist of potential presenters in the group documents.

For future workshops, we discussed setting overarching themes to focus the scope and make the workshops more thematic, in addition to dedicated time to develop a shared paper or share recent research. External speakers could be invited based on the theme. Research updates could be organized by theme or project, with detailed research discussions planned and led by designated persons. PhD students could have specific sessions with longer presentations and more detailed feedback discussions. The day before the main workshop should be reserved as a “pre-day for open activities” to facilitate informal meetings and group work.

4. Resources

- Introduction slides (2-3 slides per participant), updating on recent research, can be viewed [HERE](#)
- Collection of notes from the workshop can be viewed [HERE](#)

5. Photos from the workshop



